🌾 Agent Workflow Documentation

1. Overview

A hybrid AI system that combines:

- **ML (Linear Regression) → Predict crop yield**
- **Rules → Detect farm issues**
- **LangGraph Agent → Control workflow**
- **RAG + LLM → Generate advisory**

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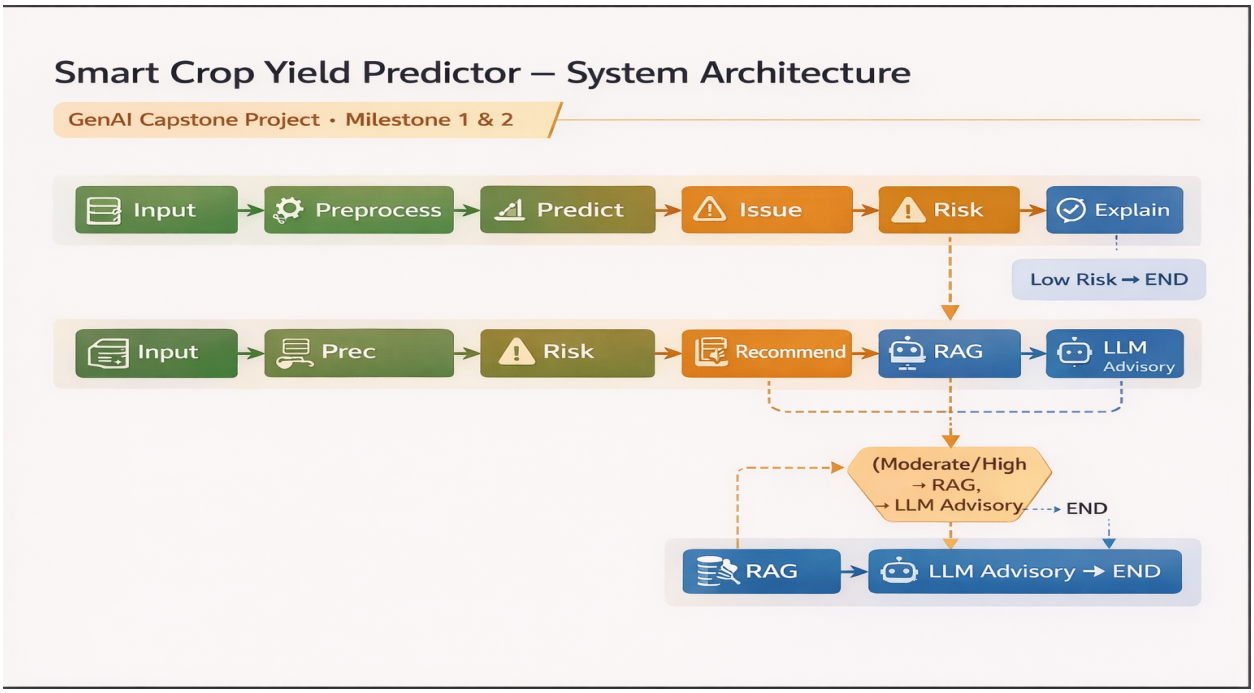
2. Workflow Pipeline



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3. Core Nodes

🔷 **Preprocess**

- One-hot encoding + scaling
- Aligns input with training columns

🔷 **Predict (ML Model)**

- Linear Regression
- Output: Predicted Yield (t/ha)

🔷 **Issue Detection (Rules)**

- Detects:
  - Low N, P, K
  - Low rainfall
  - pH imbalance
  - Heat stress

🔷 **Risk Scoring**

- Based on: { ratio = predicted_yield / crop_baseline }
- Output:
  - High (<0.6)
  - Moderate (0.6–0.85)
  - Low (≥0.85)

🔷 **Recommendation**

- Rule-based corrective actions

🔷 **Explain**

- Summarizes prediction + issues + risk
- Triggers routing logic

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4. Conditional Logic (Key Idea)

- Low Risk → Skip AI → Fast output
- Moderate/High → Use AI (RAG + LLM)

Makes system efficient + cost-optimized

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5. RAG Module

- FAISS vector DB
- MiniLM embeddings
- Top-3 relevant chunks retrieved

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6. LLM Advisory

- Model: flan-t5-large
- Generates:
  - Context-aware suggestions
  - Human-readable advisory

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7. Shared State

```
FarmState = {
  input, prediction, issues, risk,
  recommendations, rag_docs, advisory
}
```

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8. Final Output

The system outputs predicted crop yield, risk level, detected issues, and actionable recommendations. For moderate or high risk, it retrieves relevant knowledge (RAG) and generates a human-readable LLM advisory. The final result is a structured, interpretable response combining prediction, diagnosis, and intelligent guidance for better farming decisions.